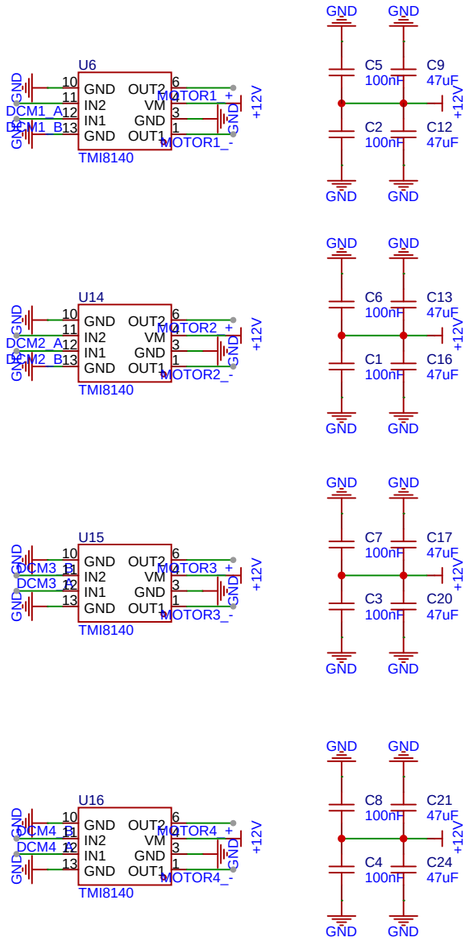
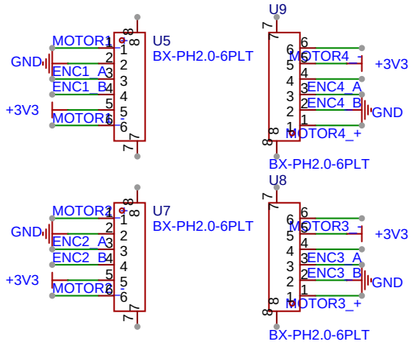
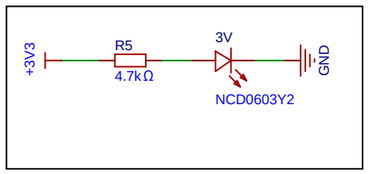
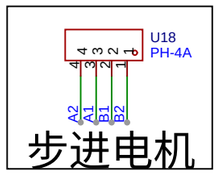

U6
GND
DCM1_A
DCM1_B
OUT2
IN2
VM
IN1
GND
OUT1
MOTOR1_+
MOTOR1_-
TMI8140
U14
DCM2_A
DCM2_B
MOTOR2_+
MOTOR2_-
U15
DCM3_A
DCM3_B
MOTOR3_+
MOTOR3_-
U16
DCM4_A
DCM4_B
MOTOR4_+
MOTOR4_-
+12V

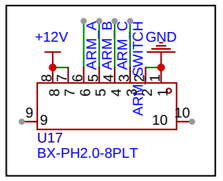
GND
C5
100nF
C9
47uF
C2
C12
C6
C13
C1
C16
C7
C17
C3
C20
C8
C21
C4
C24
+12V

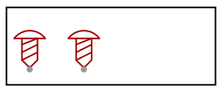

CN1
FPC0.5H1.022P
DCM4_A
ENC4_A
ARM_SWITCH
ENC4_B
DCM4_B
ARM_A
ARM_B
STEP_PUL
STEP_EN
DCM3_A
DCM3_B
ARM_C
ENC2_A
DCM1_A
ENC2_B
DCM2_A
DCM1_B
ENC3_A
DCM2_B
ENC3_B
ENC1_A
ENC1_B

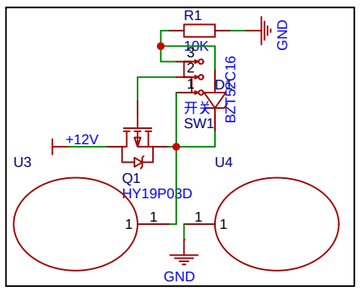

U5
BX-PH2.0-6PLT
U7
BX-PH2.0-6PLT
U9
BX-PH2.0-6PLT
U8
BX-PH2.0-6PLT
MOTOR1
MOTOR2
MOTOR3
MOTOR4
ENC1_A
ENC1_B
ENC2_A
ENC2_B
ENC3_A
ENC3_B
ENC4_A
ENC4_B
GND
+3V3

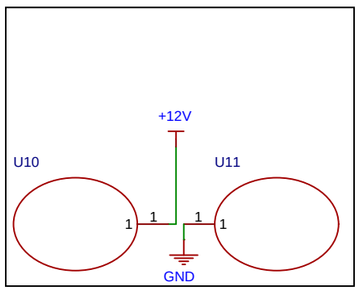

R5
4.7kΩ
3V
NCD0603Y2
+3V3
GND

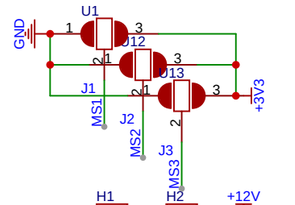

U18
PH-4A
A2
A1
B1
B2
步进电机

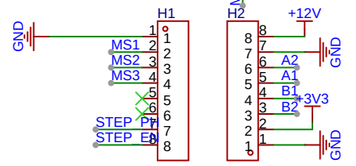

+12V
ARM_A
ARM_B
ARM_C
SWITCH
GND
U17
BX-PH2.0-8PLT

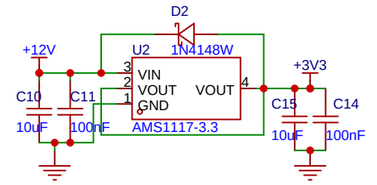



R1
10K
DC-DC16
开关
SW1
+12V
U3
U4
Q1
HY19P03D
GND


+12V
U10
U11
GND


U1
U12
U13
J1
J2
J3
MS1
MS2
MS3
+3V3
GND


H1
H2
MS1
MS2
MS3
STEP_PUL
STEP_EN
A2
A1
B1
B2
+12V
+3V3
GND


D2
1N4148W
U2
VIN
VOUT
GND
AMS1117-3.3
C10
C11
C15
C14
10uF
100nF
+12V
+3V3

舵机